

Kai Li, PhD

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RESEARCH INTERESTS Scholarly communication; Quantitative science studies; Digital humanities; Information organization; Science and technology studies; Information visualization

ACADEMIC POSITION *Assistant Professor* 2023 -
School of Information Sciences
University of Tennessee, Knoxville, Tennessee, USA

Assistant Professor 2020 - 2022
School of Information Resource Management
Renmin University of China, Beijing, China

EDUCATION *Ph.D. in Information Studies* 2015 - 2020
College of Computing & Informatics
Drexel University, Philadelphia, PA
• Dissertation: A quantitative examination of research methods and software in scientific publications
• Advisor: Dr. Erjia Yan

Master of Science in Library and Information Science 2012 - 2013
School of Information Studies
Syracuse University, Syracuse, NY

Bachelor in History 2003 - 2007
School of History
Beijing Normal University, Beijing, China

PUBLICATIONS Peer-Reviewed Journal Articles

[J26] Li, K., Dobreski, B. & Busch, M. (2024). How are library cataloging metadata used differently over time: a large-scale quantitative analysis of the Library of Congress catalog. *Journal of Documentation*.

[J25] Li, K., Zheng, X. & Ni, C. (2024). The Anglicization of science in China. *Research Evaluation*, Volume 33, rvae036, <https://doi.org/10.1093/reseval/rvae036>.

[J24] Li, K., Ma, R., & Fang, Z. (2024). Explicit or Implicit Digital Humanities? An Examination of Search Strategies to Retrieve Digital Humanities Publications from Large-Scale Scholarly Databases. *Quantitative Science Studies*, 1-23. https://doi.org/10.1162/qss_a_00321

[J23] Wang, Y., & Li, K. (2024). How do official software citation formats evolve over time? A longitudinal analysis of R programming language packages. *Scientometrics*, 129, 3997-4019. <https://doi.org/10.1007/s11192-024-05064-6>

[J22] Jiao, C., Li, K., & Fang, Z. (2023). How are exclusively data journals indexed

in major scholarly databases? An examination of four databases. *Scientific Data*, 10(1), 737. <https://doi.org/10.1038/s41597-023-02625-x>

[J21] Jiao, C., Li, K., & Fang, Z. (2022). Data sharing practices across knowledge domains: a dynamic examination of data availability statements in PLOS ONE publications *Journal of Information Science*, 50(3), 673-689. <https://doi.org/10.1177/0165551522110183>

[J20] Li, K. (2022). Relationships between Method-section citation rates and citation contexts: Evidence from highly cited references in psychology. *Online Information Review*, 46(5): 829-845 <https://doi.org/10.1108/OIR-03-2021-0171>

[J19] Wang Y., Zhang C., Li K. (2022) A review on method entities in the academic literature: extraction, evaluation, and application. *Scientometrics*. 127: 2479-2520. <https://doi.org/10.1007/s11192-022-04332-7>

[J18] Ma, R., & Li, K. (2022). Digital humanities as a cross-disciplinary battleground: An examination of inscriptions in journal publications. *Journal of the Association for Information Science and Technology*, 73(2): 172-187. <https://doi.org/10.1002/asi.24534>

[J17] Li, K. & Jiao, C. (2022). The data paper as a socio-linguistic epistemic object: A content analysis on the rhetorical moves used in data paper abstracts. *Journal of the Association for Information Science and Technology*, 73(6): 834-846. <https://doi.org/10.1002/asi.24585>

[J16] Li, K., Jiao, C., Sugimoto, C. R., & Lariviere, V. (2022). Versioning boundary objects: the citation profile of the Diagnostic and Statistical Manual for Mental Disorders (DSM). *Journal of Documentation*, 78(4): 871-889. <https://doi.org/10.1108/JD-06-2021-0117>

[J15] Lou, W. Su, Z., He, J. & Li, K. (2021) A temporally dynamic examination of research method usage in the Chinese library and information science community. *Information Processing & Management*, 58(5): 102686. <https://doi.org/10.1016/j.ipm.2021.102686>

[J14] Wu C., Yan, E., Zhu, Y., & Li, K. (2021). Gender imbalance in the productivity of funded projects: A study of the outputs of National Institutes of Health R01 grants. *Journal of the Association for Information Science and Technology*, 72(11): 1386-1399. <https://doi.org/10.1002/asi.24487>

[J13] Li, K. (2021) The re-instrumentalization of the Diagnostic and Statistical Manual of Mental Disorders (DSM) in psychological publications: a citation context analysis. *Quantitative Science Studies*, 2(2): 678-697. https://doi.org/10.1162/qss_a_00124

[J12] Lou, W., Zhang, J., Li, K., & He, J. (2020). Understanding the Application of Science Mapping Tools in LIS and Non-LIS Domains. *Data and Information Management*, 4(2), 94-108. <https://doi.org/10.2478/dim-2020-0006>

[J11] Yan, E., Chen, Z. & Li, K. (2020). The relationship between journal citation impact and citation sentiment: A study of 32 million citances in PubMed Central, *Quantitative Science Studies*, 1(2): 664-674. https://doi.org/10.1162/qss_a_00040

- [J10] Yan, E., Chen, Z. & Li, K. (2020). Authors' status and the perceived quality of their work: Measuring citation sentiment change in nobel articles. *Journal of the Association for Information Science and Technology*, 71(3): 314-324. <https://doi.org/10.1002/asi.24237>
- [J9] Li, K., Greenberg, J. and Dunic, J. (2020). Data objects and documenting scientific processes: An analysis of data events in biodiversity data papers. *Journal of the Association for Information Science and Technology*, 71(2): 172-182. <https://doi.org/10.1002/asi.24226>
- [J8] Li, K., Chen, P.-Y., & Yan, E. (2019). Challenges of measuring software impact through citations: An examination of the lme4 R package. *Journal of Informetrics*, 13(1), 449-461. <https://doi.org/10.1016/j.joi.2019.02.007>
- [J7] Li, K., & Yan, E. (2019). Are NIH-funded publications fulfilling the proposed research? An examination of concept-matchedness between NIH research grants and their supported publications. *Journal of Informetrics*, 13(1), 226-237. <https://doi.org/10.1016/j.joi.2019.01.001>
- [J6] Li, Y., Wu, C., Yan, E., & Li, K. (2018). Will open access increase journal CiteScores? An empirical investigation over multiple disciplines. *PLOS ONE*, 13(8): e0201885. <https://doi.org/10.1371/journal.pone.0201885>
- [J5] Yan, E., & Li, K. (2018). Which domains do open-access journals do best in? A 5-year longitudinal study. *Journal of the Association for Information Science and Technology*, 69(6): 844-856. <https://doi.org/10.1002/asi.24002>
- [J4] Li, K., Rollins, J., & Yan, E. (2018). Web of Science use in published research and review papers 1997–2017: a selective, dynamic, cross-domain, content-based analysis. *Scientometrics*, 115(1), 1-20. <https://doi.org/10.1007/s11192-017-2622-5>
- [J3] Zhao, M., Yan, E., & Li, K. (2018). Data set mentions and citations: A content analysis of full-text publications. *Journal of the Association for Information Science and Technology*, 69(1): 32-46. <https://doi.org/10.1002/asi.23919>
- [J2] Li, K. & Yan, E. (2018). Co-mention network of R packages: Scientific impact and clustering structure. *Journal of Informetrics*, 12(1): 87-100. <https://doi.org/10.1016/j.joi.2017.12.001>
- [J1] Li, K., Yan, E., & Feng, Y. (2017). How is R cited in research outputs? Structure, impacts, and citation standard. *Journal of Informetrics*, 11(4): 989-1002. <https://doi.org/10.1016/j.joi.2017.08.003>

Peer-reviewed Full Conference Papers

- [C5] Istrate, A., Fisher, J., Yang, X., Moraw, K., Li, K., & Li, D. (2024). Scientific Software Citation Intent using Large Language Models. 1st Workshop on Natural Scientific Language Processing and Research Knowledge Graphs (NSLP 2024).
- [C4] Li, K., & Yan, E. (2019). Using a keyword extraction pipeline to understand concepts in future work sections of research papers. In *17th International Conference on Scientometrics and Informetrics*.
- [C3] Feng, Y., Li, K., & Agosto, D.E. (2017). Healthy users' personal health in-

formation management from activity trackers: The perspective of gym-goers. In *Proceedings of the Annual Meeting of Association for Information Science and Technology* (54: pp. 71-81). <https://doi.org/10.1002/pra2.2017.14505401009>

[C2] Li, K., Greenberg, J., & Lin, X. (2016). Software Citation, Reuse and Metadata Considerations: An Exploratory Study Examining LAMMPS. In *Proceedings of the Annual Meeting of the Association for Information Science and Technology* (53: pp. 1-10). <https://doi.org/10.1002/pra2.2016.14505301072>

[C1] Qin, J., & Li, K. (2013). How portable are the metadata standards for scientific data? a proposal for a metadata infrastructure. In *Proceedings of the International Conference on Dublin Core and Metadata Applications* (pp. 25-34). <https://dcpapers.dublincore.org/pubs/article/view/3670>

Refereed Short Papers in Conference Proceedings

[S8] Li, K., Huang, P. P., & Jeng, W. (2025). Are data papers cited as research data? Preliminary analysis on interdisciplinary data paper citations. In *Proceedings of iConference 2025*.

[S7] Cui, H., Wang, Y., & Li, K. (2025). Beyond Citations: Tracing and Validating the Rapid Adoption of AlphaFold in Biomedical Research Through Full-Text Analysis. In *20th International Conference on Scientometrics and Informetrics*.

[S6] Li, K., Chen, J., & Ni, C. (2023). The patterns of publication languages used by STEM research in China. In *19th International Conference on Scientometrics and Informetrics*.

[S5] Li, K. & Jiao, C. (2021). How are data paper abstracts constructed? Preliminary analysis of rhetorical moves in data paper abstracts from Scientific Data and Data in Brief. In *18th International Conference on Scientometrics and Informetrics*.

[S4] Dagiene, E. & Li, K. (2021). ISBNs as identifiers for books in research evaluations. In *18th International Conference on Scientometrics and Informetrics*.

[S3] Ma, R., Li, K., & He, D. (2021). Understanding the Narrative Functions of Visualization in Digital Humanities Publications: A Case Study of the Journal of Cultural Analytics. In *Proceedings of iConference 2021*.

[S2] Pascua, S., Li, K., & Greenberg, J. (2019). Toward A Metadata Activity Matrix: Conceptualizing and Grounding the Research Life-cycle and Metadata Connections. In *Proceedings of the International Conference on Dublin Core and Metadata Applications*.

[S1] Guerra, J., Wei, Q., Li, K., Ahumada, L., Winston, F., & Desai, B. R. (2018). Scosy: A Biomedical Collaboration Recommendation System. Presented at the 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Honolulu, HI, USA.

Refereed Conference Posters

[P10] Li, K., Ma, R.Q., & Fang Z.C. (2023). An examination of different strategies to retrieve digital humanities publications from large-scale scientific databases. Presented in *19th International Conference on Scientometrics and Informetrics*.

[P9] Li, K., Zheng, X., & Ni, C. (2023). Gender disparities in the STEM research enterprise of China. 2nd International Conference of Science of Science & Innovation, Evanston, IL, USA.

[P8] Chen, PY, Li, K. & Jiao, C. (2022), A Preliminary Analysis of Geography of Collaboration in Data Papers by S&T Capacity Index. *Proceedings of the Association for Information Science and Technology*, 59: 642-644.

[P7] Ma, R.Q. & Li, K. (2020). Telling Multifaceted Stories with Humanities Data: Visualizing Book of Hours Manuscripts. Presented at iConference 2020 Annual Meeting. (Finalist for Best Poster Award).

[P6] Li, K., Chen, P.-Y., & Fang, Z. (2019). Disciplinarity of Software Papers: A Preliminary Analysis. In *Proceedings of the Annual Meeting of Association for Information Science and Technology*.

[P5] Li, K., & Chen, P.-Y. (2019). A preliminary scientometric analysis of the Cross-Strait scientific collaboration. In *17th International Conference on Scientometrics and Informetrics*.

[P4] He, J. & Li, K. (2019). How comprehensive is the PubMed Central Open Access full-text database?. Presented at iConference 2019 Annual Meeting, Washington, DC, USA. <http://hdl.handle.net/2142/103317>

[P3] Li, K. & Chen, PY. (2018). The narrative structure as a citation context in data papers: A preliminary analysis of Scientific Data. In *Proceedings of the Annual Meeting of Association for Information Science and Technology*. <https://doi.org/10.1002/pr2.2018.14505501147>

[P2] Li, K. & Xu, S. (2017). Measuring the impact of R packages. In *Proceedings of the Annual Meeting of Association for Information Science and Technology* (54: pp. 739-741). <https://doi.org/10.1002/pr2.2017.14505401138>

[P1] Li, K. (2016). How much can data citation standards be used for scientific software? A crosswalk analysis of data citation standards for software citation needs. Presented in Research Data Alliance 8th Plenary Meeting, Denver, Colorado, September 14-17, 2016.

Book Reviews

[B1] Li, K. (2019). A review of The Politics of Mass Digitization. *Education for Information*, 35, 361-363. <https://doi.org/10.3233/EFI-199011>

Doctoral Colloquia (Lightly Peer-Reviewed)

[D2] Li, K. (2019). Quantitative examination of research methods and software in scientific publications. Selected to attend the Doctoral Colloquium at ASIS&T 2019, Melbourne, Australia, October 19, 2019.

[D1] Li, K. (2019). What is the disciplinarity of software. Selected to attend the Doctoral Forum at 17th International Conference on Scientometrics and Informetrics, Rome, Italy, September 2, 2019.

Non-Refereed Conference Presentations

[T1] Li, K. (2012). RDA in China. Presented at ALA Annual Conference 2012. June 24, 2012. Anaheim, California.

TEACHING EXPERIENCE

Instructor

School of Information Sciences, University of Tennessee, Knoxville

- **INSC 512: Information Organization & Retrieval**
- **INSC 592: Introduction to Data Analytics and Visualization**

School of Information Resource Management, Renmin University of China

- **Information Visualization in the Context of Digital Humanities** (Master level) (Fall '21, '22)
- **Information Visualization** (Undergraduate level) (Spring '22)
- **Knowledge Organization** (Master level) (Spring '22)

College of Computing & Informatics, Drexel University

- **INFO 250: Information Visualization** (Online) (Summer '20)
- **INFO 590: Foundations of Data and Information** (Online) (Winter '20)
- **INFO 250: Information Visualization** (On-site) (Summer '19)

HONORS & AWARDS

Bonnie Carroll and Roy Cooper Faculty Enrichment Award 2024
School of Information Sciences, University of Tennessee, Knoxville

- Awarded \$2,500

IAS Collaborative Award 2023-2024
Indiana University

- Co-PI
- Project: Empirical Exploration of the Visual Language in Digital Humanities
- Awarded \$6,000

LIS Education and Data Science for the National Digital Platform (LEADS-4-NDP) 2019
Fellowship

Drexel University

- Project: Automatic Identification of Publisher Entities to Support Discovery and Navigation (with OCLC)
- Awarded \$5,000 plus travel support

RDA/US Data Share Fellowship 2016 - 2017
Research Data Alliance

- Project: Metadata Standards and Software Citation Practices
- Awarded \$10,000 plus travel support

Student Travel Award 2019
The International Society for Informetrics and Scientometrics

**SERVICE &
MEMBERSHIPS**

Editorial Board

- **Nature Scientific Data**, 2024-Present
- **Nature Humanities & Social Sciences Communications**, 2024-Present
- **Annual Review of Information Science and Technology (ARIST)**, 2022-2024

Service

- **Committee Member**, Dublin Core Metadata Initiative (DCMI) Annual Conference 2024, Student Forum
- **President**, ASIS&T SIG-MET, 2023-2024
- **President-Elect**, ASIS&T SIG-MET, 2022-2023
- **Jury**, ASIS&T Lois Lunin Award, 2022, 2023
- **Mentor**, ASIS&T Mentor Program, 2022-2023
- **Webmaster**, ASIS&T SIG-MET, 2021-2022
- **Committee Member**, ISSI 2021 Paper of the Year Award Committee, 2021
- **Secretary**, ASIS&T SIG-MET, 2020-2021
- **Award Coordinator**, ASIS&T SIG-MET, 2019-2020

Selected Peer Review Experience: (My full peer review record can be retrieved from my ORCID page.)

- Journal of the Association for Information Science and Technology
- Humanities and Social Sciences Communications
- Scientific Reports
- Quantitative Science Studies
- Scientometrics
- Journal of Informetrics
- Journal of Information Science
- PLoS ONE
- ASIS&T Annual Conference
- iConference

Membership

- The Association for Information Science and Technology (ASIS&T)

**PROFESSIONAL
EXPERIENCE**

Cataloger

2014 - 2015

Ingram Content Group, Fort Wayne, Indiana

Cataloging Librarian

2007 - 2012

Capital Library of China, Beijing, China